

```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[GH_FFCutsFits,CutFitFileGH] =
recover_selected_cuts('GH_FFCutsFits','e:\SHARE\FE\Data\analysis\07_fit_with_multicu
t_all\sel_cuts');
```

```
fn = fieldnames(GH_FFCutsFits);
%nf = 11
%for nf = 2:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei400_off110GH001minFF';
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('*****\n');
```

```
*****
```

```
fprintf('Processing cuts: %s \n', selectCuts);
```

```
Processing cuts: Ei400_off110GH001minFF
```

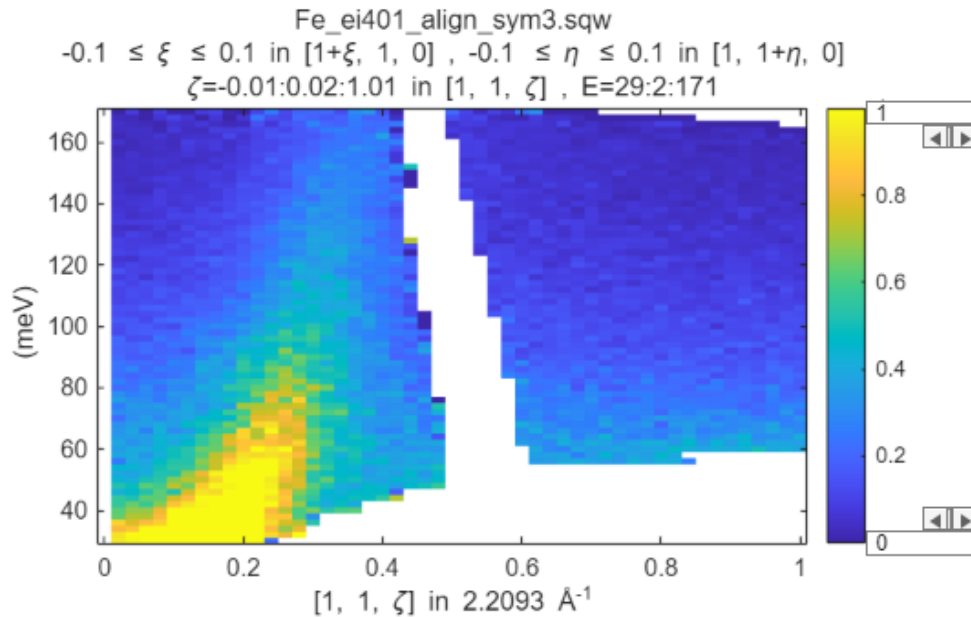
```
fprintf('*****\n');
```

```
*****
```

```
cuts_list = GH_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "GH";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

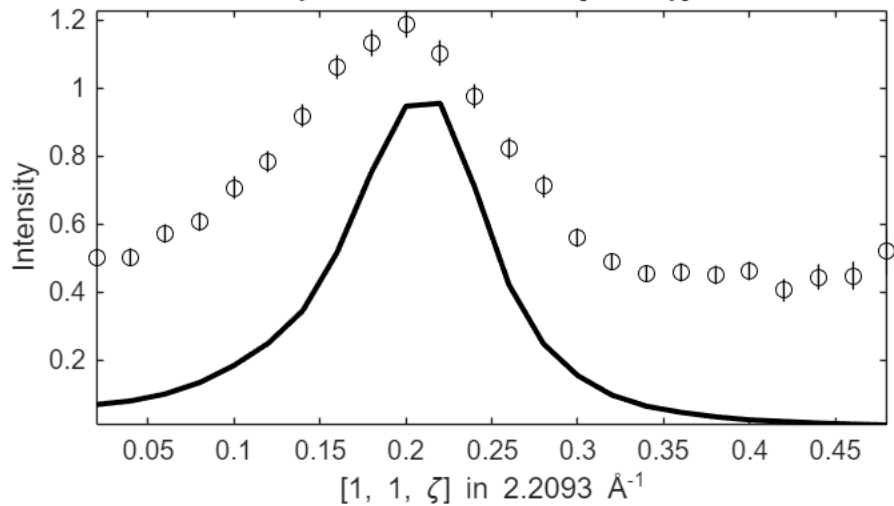
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GH_FFCutsFits.(selectCuts) = cuts_list;

GH_FFCutsFits = fit_multicutsFF_along_direction(...
    GH_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

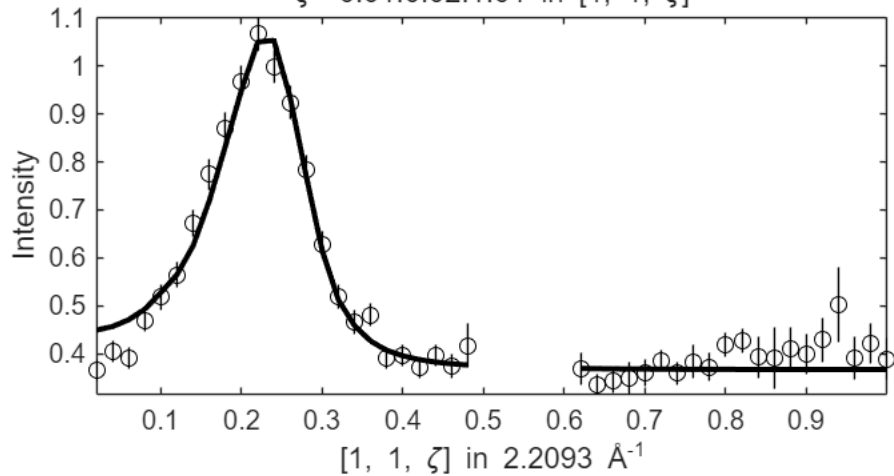
```
*****
**** En = 45±5 going to 165
*****
```

Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $40 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[1, 1, \zeta]$



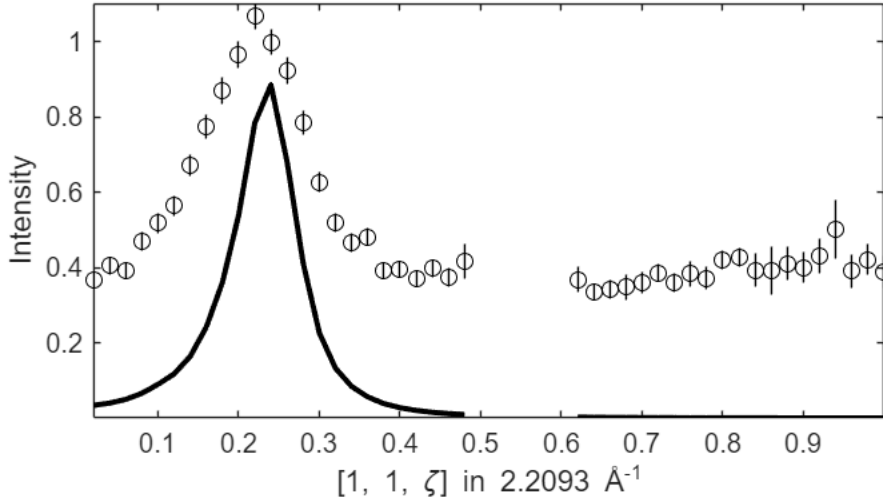
 **** En = 55±5 going to 165

Fe_ei401_align_sym3.sqw
 $S=0.87\pm0.062$; $J_0= 25\pm0.47$; $J_1= 0\pm 0$; $\gamma=16.8318\pm2.21112$
 $0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $50 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

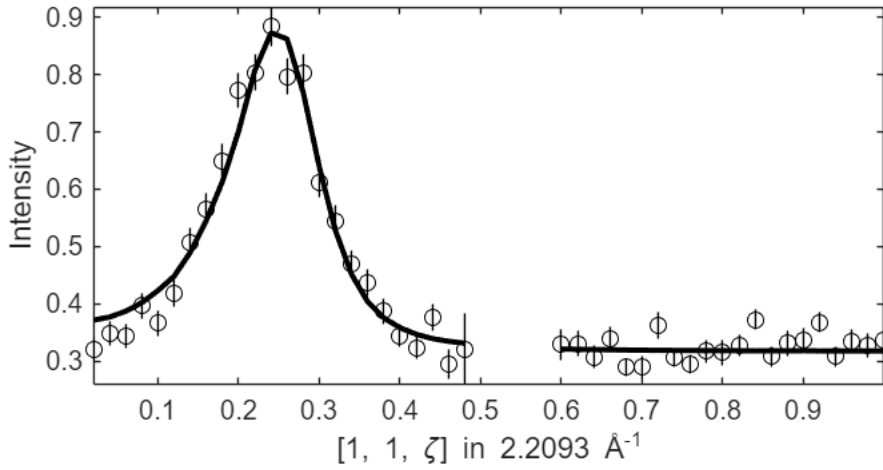
$0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $50 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[1, 1, \zeta]$



 **** En = 65±5 going to 165

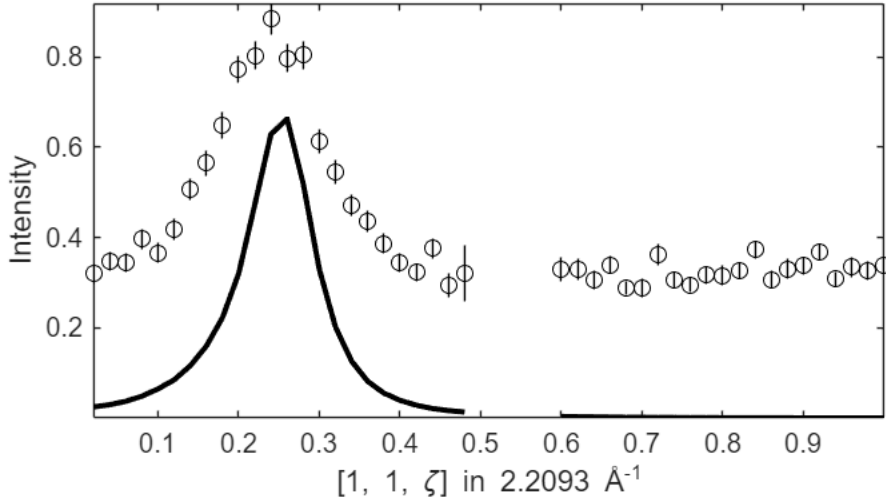
Fe_ei401_align_sym3.sqw

$S=0.86\pm0.025$; $J_0= 26\pm0.45$; $J_1= 0\pm 0$; $\gamma=22.3834\pm0.655974$
 $0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $60 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

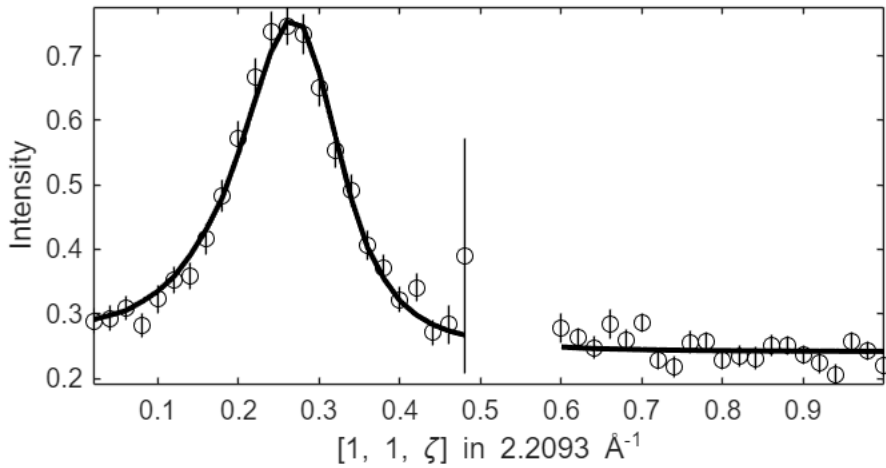
$0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $60 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[1, 1, \zeta]$



 **** En = 75±5 going to 165

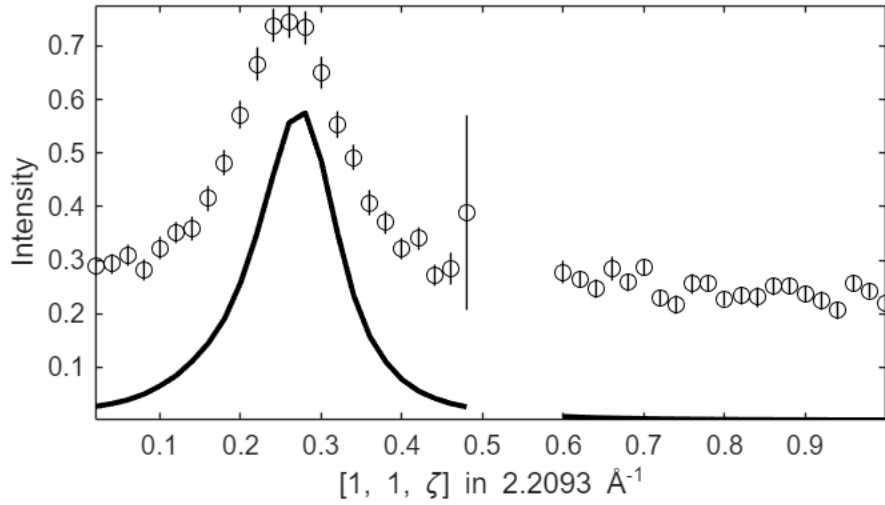
Fe_ei401_align_sym3.sqw

S= 1.1±0.062; J0= 27±0.54; J1= 0± 0; gamma=32.6156±3.15329
 $0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $70 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

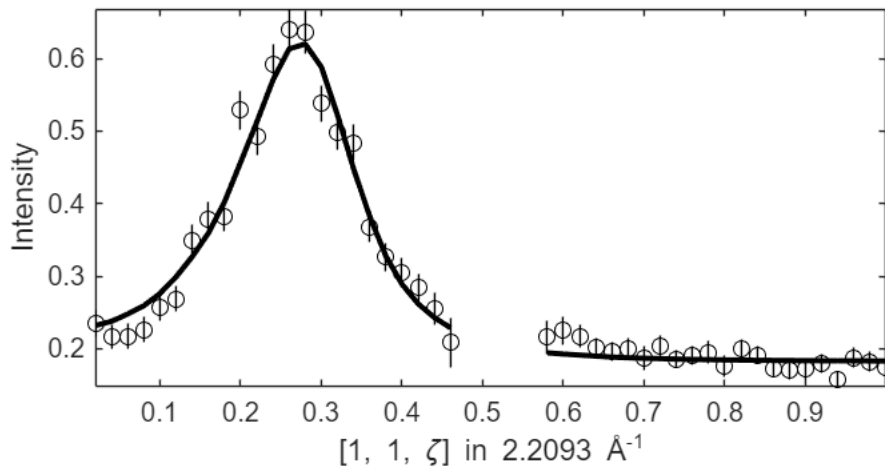
$0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $70 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[1, 1, \zeta]$



 **** En = 85±5 going to 165

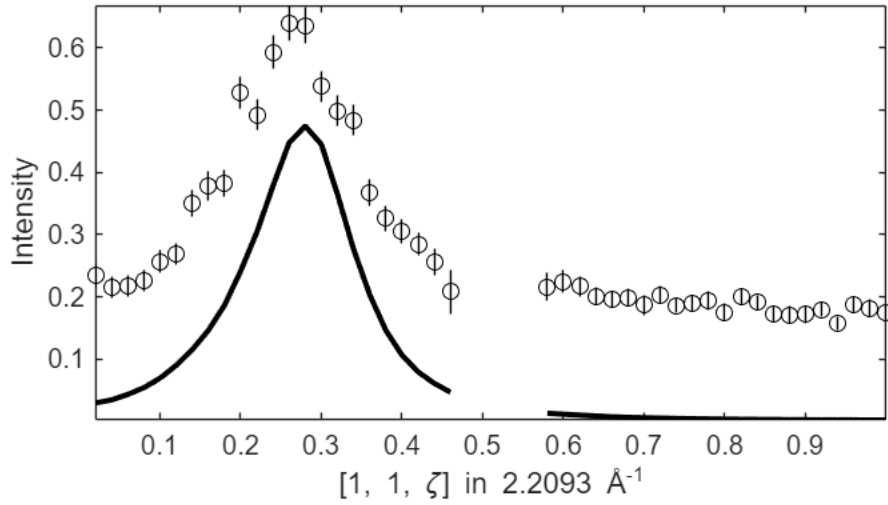
Fe_ei401_align_sym3.sqw

S= 1.3±0.03; J0= 31±0.44; J1= 0± 0; gamma=51.1368±1.03513
 $0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $80 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

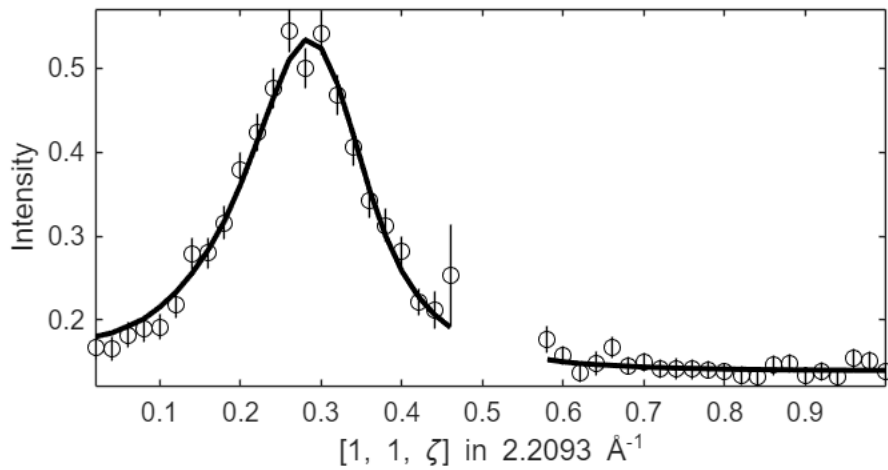
$0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $80 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[1, 1, \zeta]$



 **** En = 95±5 going to 165

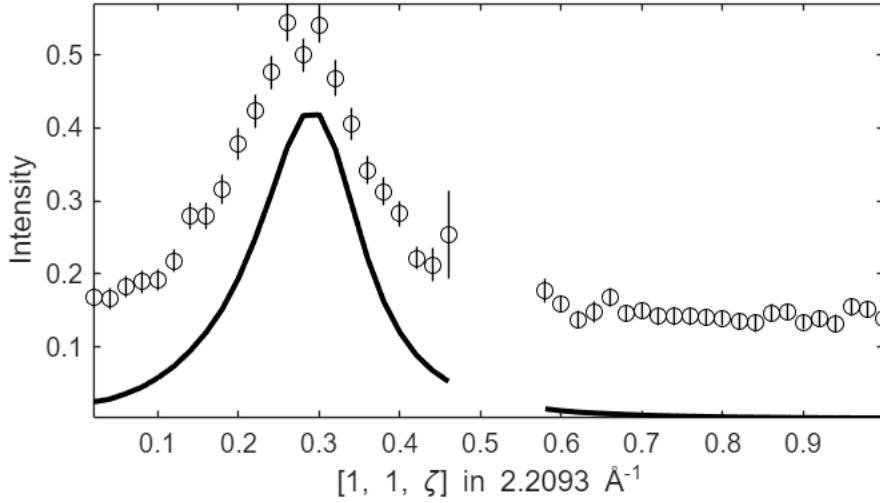
Fe_ei401_align_sym3.sqw

S= 1.3±0.072; J0= 33±0.72; J1= 0± 0; gamma=56.9138±4.95782
 $0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $90 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

$0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $90 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[1, 1, \zeta]$

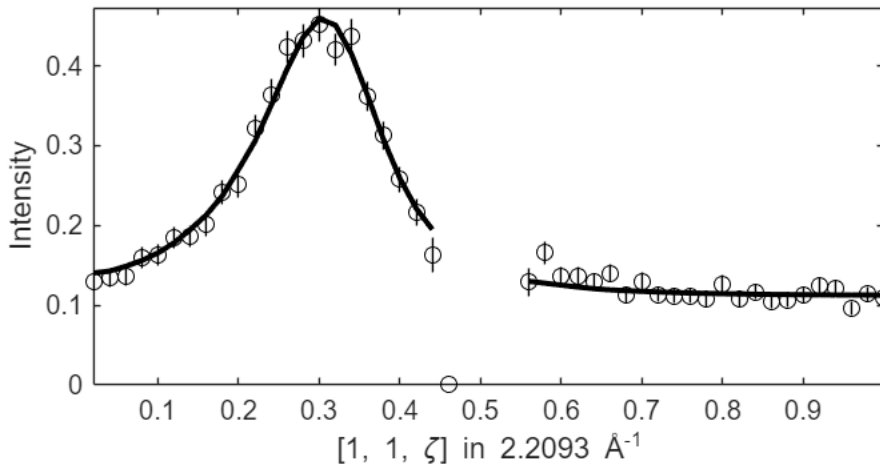


**** En = 105±5 going to 165

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9608

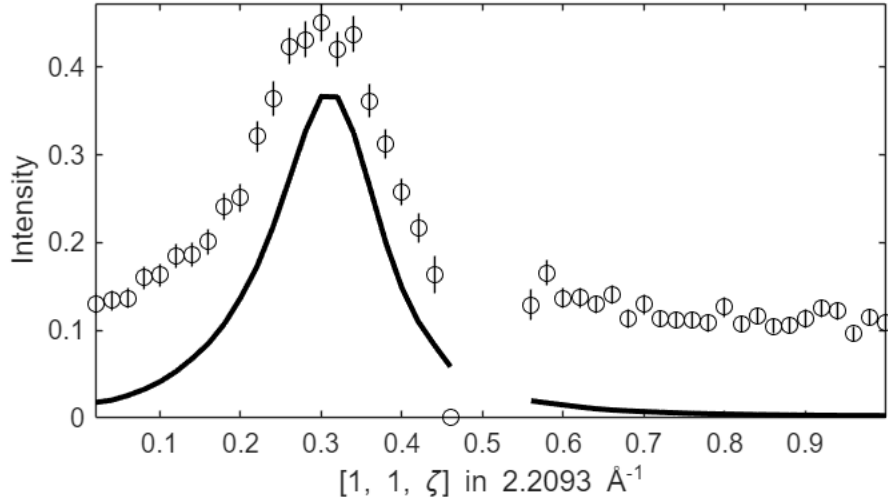
Fe_ei401_align_sym3.sqw

S= 1.2±0.022; J0= 32±0.35; J1= 0± 0; gamma=58.2315±0.927246
 $0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $100 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

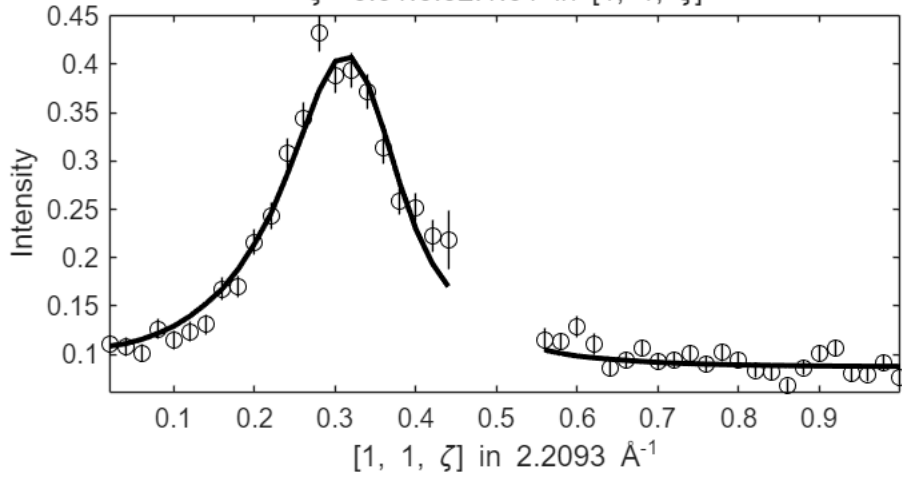
$.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $100 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[1, 1, \zeta]$



 **** En = 115±5 going to 165

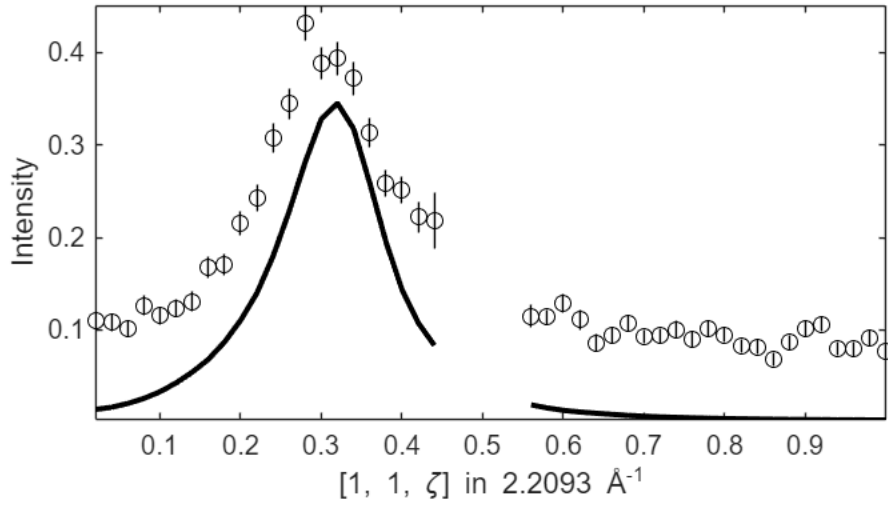
Fe_ei401_align_sym3.sqw

S= 1.1±0.078; J0= 33±0.76; J1= 0± 0; gamma=56.7747±6.41655
 $.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $110 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

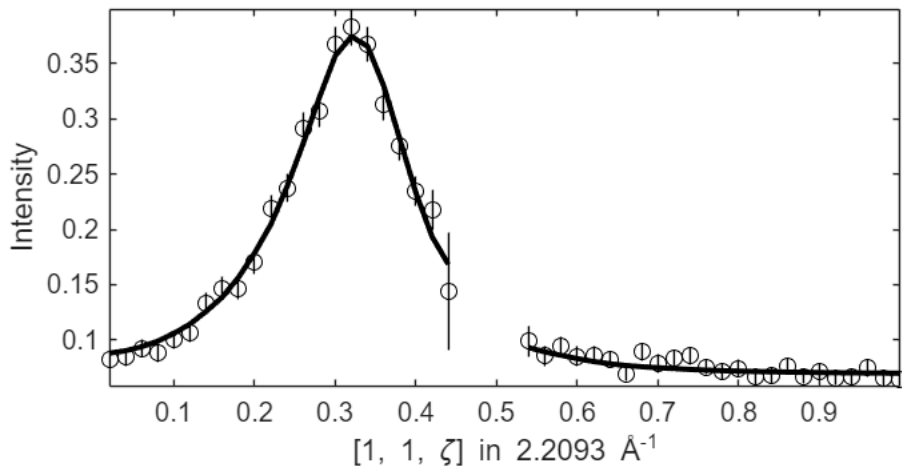
$.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $110 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[1, 1, \zeta]$



 **** En = 125±5 going to 165

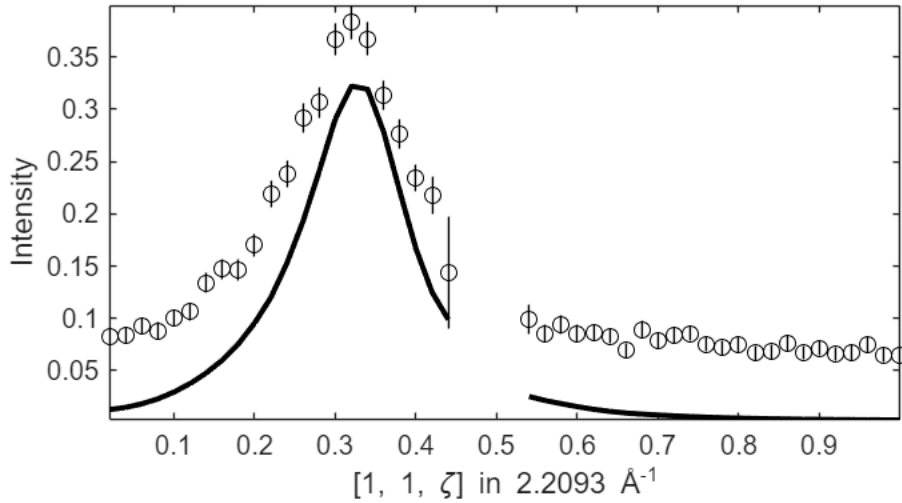
Fe_ei401_align_sym3.sqw

S= 1.1±0.017; J0= 34±0.28; J1= 0± 0; gamma=61.5448±0.793468
 $.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $120 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

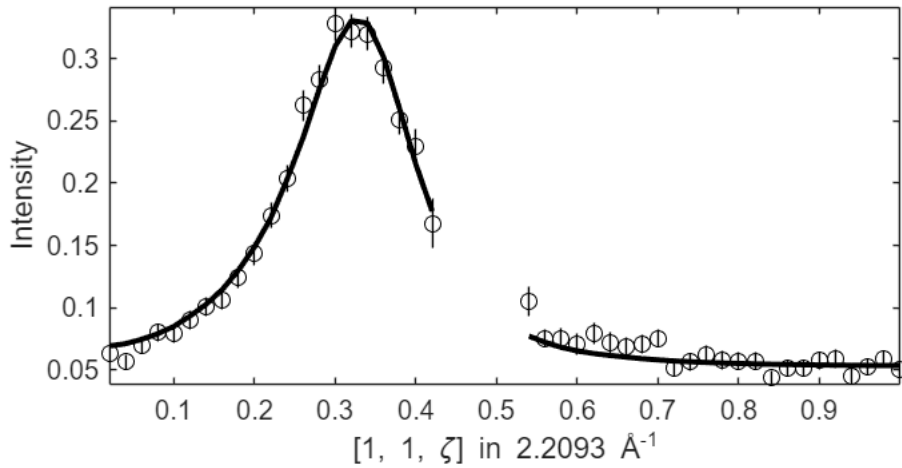
$.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $120 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[1, 1, \zeta]$



 **** En = 135±5 going to 165

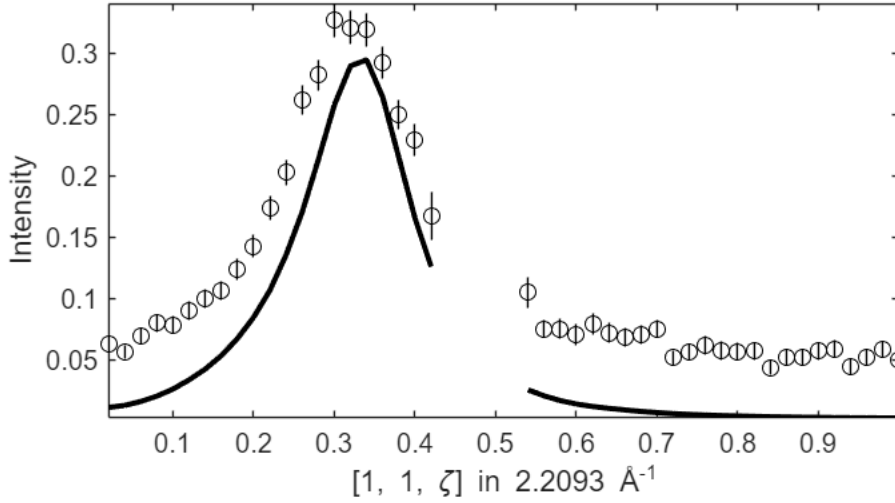
Fe_ei401_align_sym3.sqw

S= 1.1±0.061; J0= 36±0.55; J1= 0± 0; gamma=67.099±5.52461
 $.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $130 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

$.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $130 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[1, 1, \zeta]$



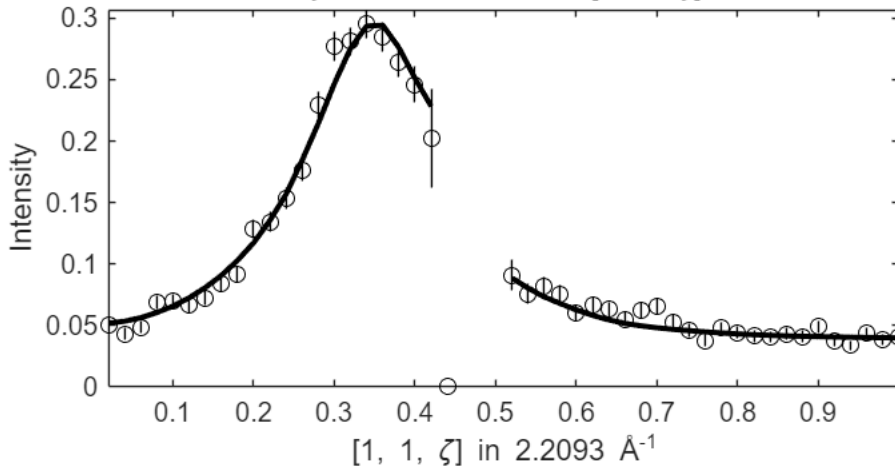
 **** En = 145±5 going to 165

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9608

Fe_ei401_align_sym3.sqw

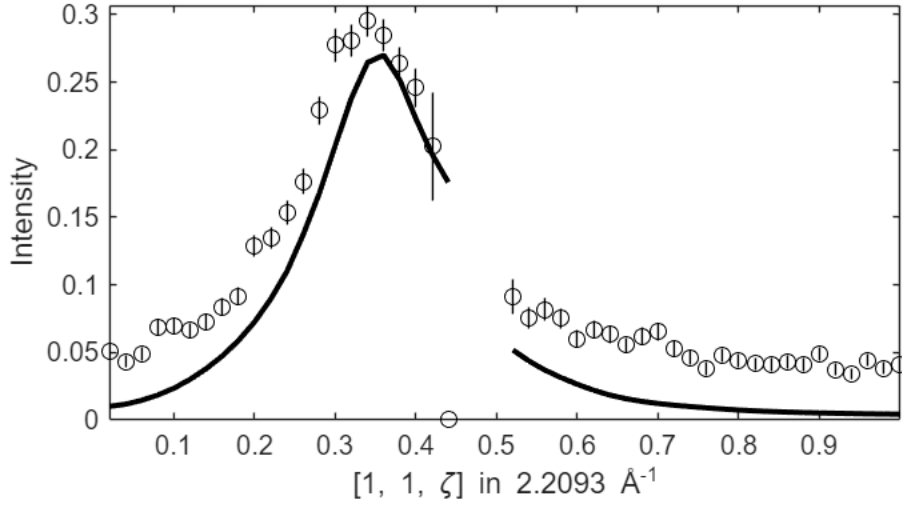
S= 1.2±0.035; J0= 35± 0.5; J1= 0± 0; gamma=82.5381±1.92007

$.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $140 \leq E \leq$
 $\zeta=-0.01:0.02:1.01$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

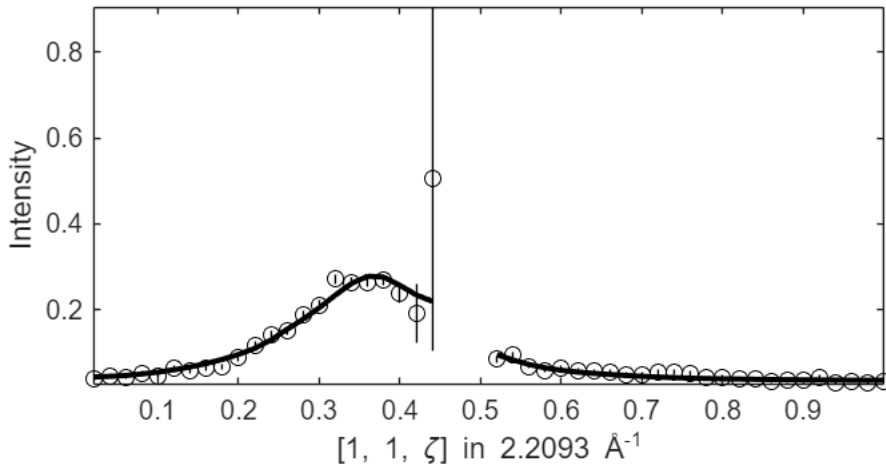
$.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $140 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[1, 1, \zeta]$



 **** En = 155±5 going to 165

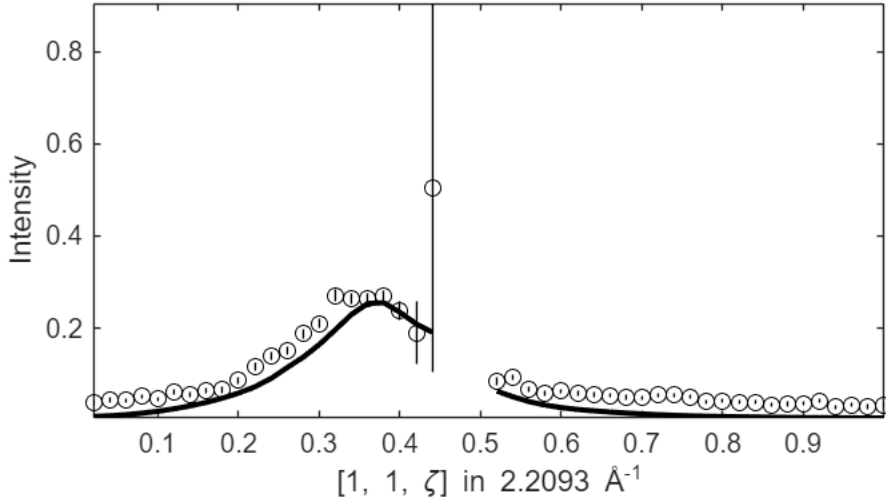
Fe_ei401_align_sym3.sqw

S= 1.2±0.043; J0= 35±0.45; J1= 0± 0; gamma=83.0904±2.40835
 $.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $150 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

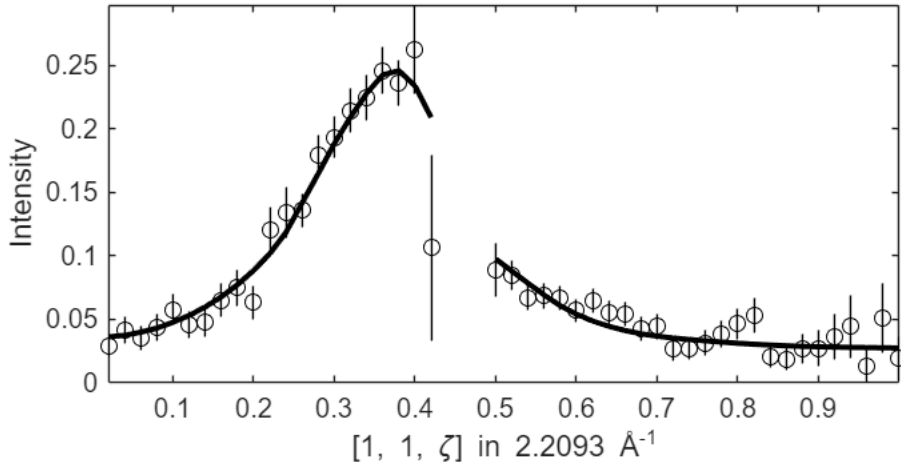
$.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $150 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[1, 1, \zeta]$



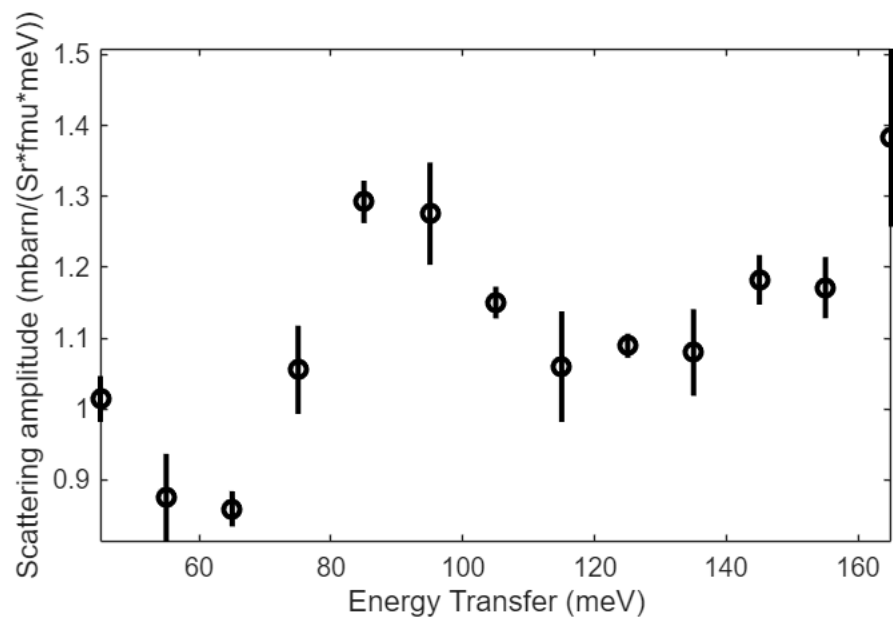
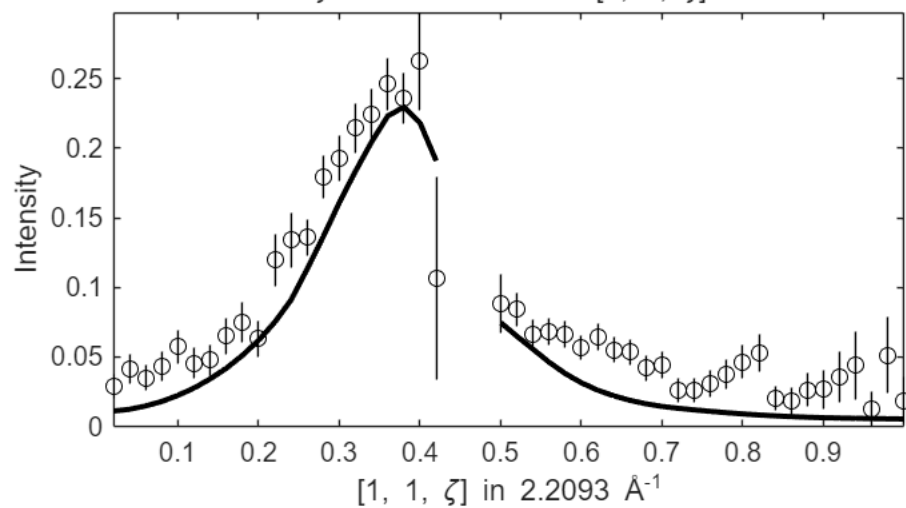
 **** En = 165±5 going to 165

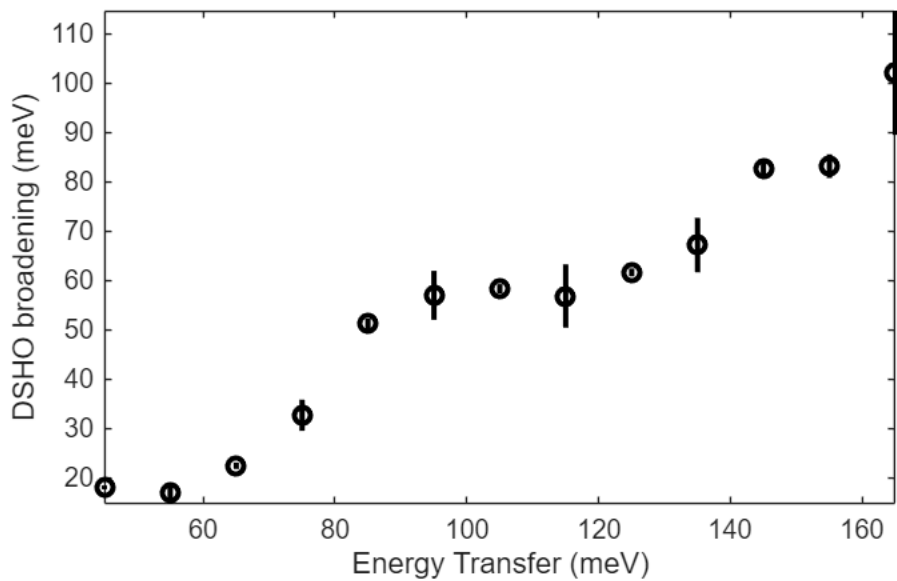
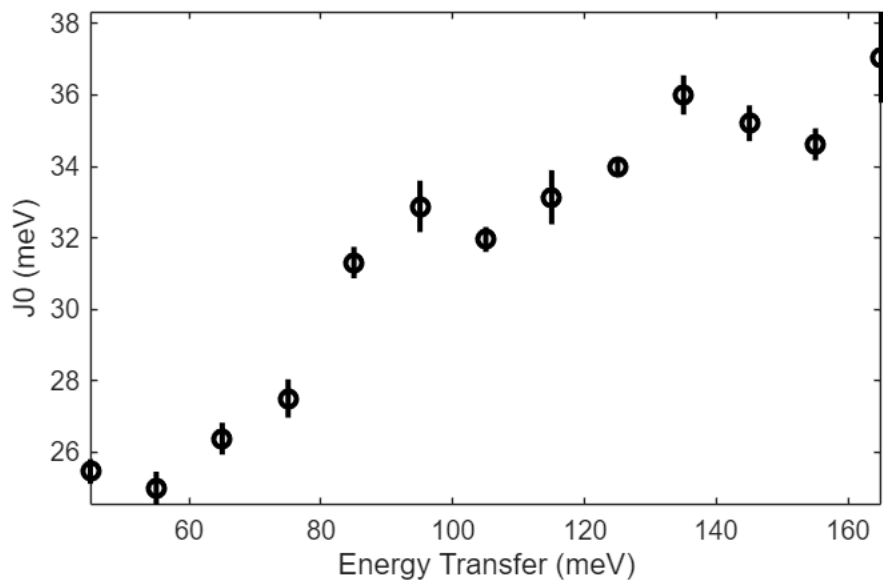
Fe_ei401_align_sym3.sqw

S= 1.4±0.13; J0= 37± 1.3; J1= 0± 0; gamma=102.023±12.5811
 $.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $160 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw
 $.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $160 \leq E \leq$
 $\zeta = -0.01:0.02:1.01$ in $[1, 1, \zeta]$





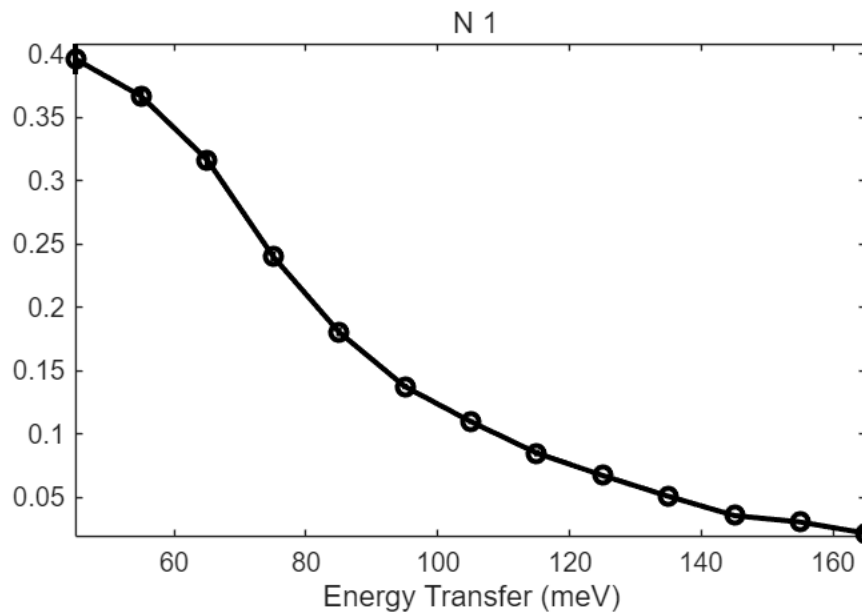
```

c1 = GH_FFCutsFits.(selectCuts);
GH_FFCutsFits.(selectCuts) = [c1{:}];

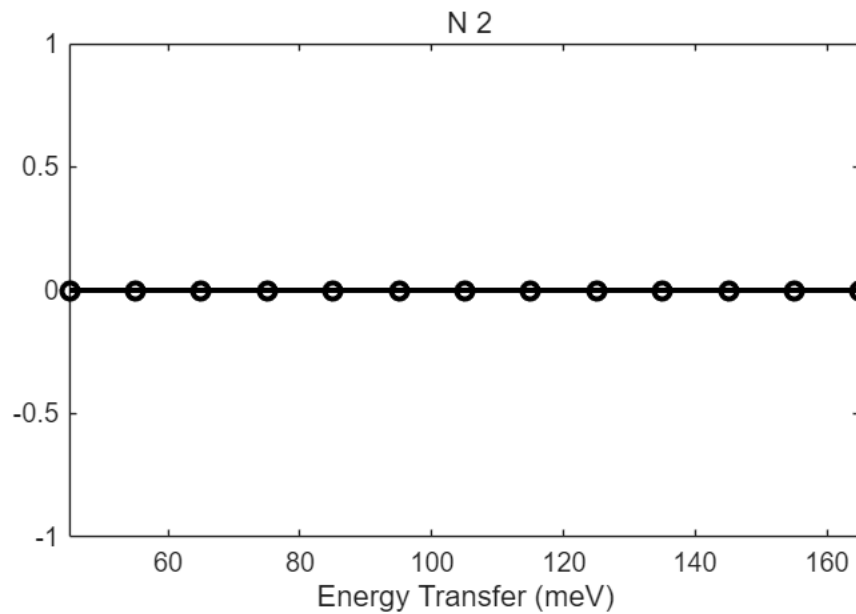
beep
beep
beep

all_fit_par = GH_FFCutsFits.([selectCuts,'_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GH.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GH_',selectCuts,'.pdf']);
export(thisFile,targ_file);
```

```
fit_src_struct = GH_FFCutsFits;  
save(CutFitFileGH,"fit_src_struct",-v7.3');  
close all;  
%end
```